Experiment Report: Memory vs. JEPA World Model — Dense World, Scarce (No Reposition)

Experiment ID: `20260717_memory_vs_wm_dense_scarce` **Date:** 2026-07-27 **Trials:** 5 trials × 5 conditions × 10 creatures = **250 creatures analyzed Analysis script:** `analysis/experiments/20260717_memory_vs_wm_dense_scarce.py` **Data:** `ml/data_20260717_memory_vs_wm_dense_scarce/`

Purpose

`20260714_memory_vs_wm_dense_reposition` found that every strategic advantage seen in the original `20260709_memory_vs_wm_v1` experiment (JEPA survival edge, memory-filter survival penalty, JEPA Tedium suppression) vanished once food became abundant and self-replenishing (`reposition=true`). This experiment restores scarcity (`reposition=false`, same as the original) while keeping everything else about the dense-world setup — 1200×900 world, 10 creatures, the same food/hazard object counts — to test whether scarcity alone brings those effects back, run on CCAD where JEPA inference overhead is negligible (~6.5ms mean, vs. the original local run's ~48ms).

Note on world "density": the world dimensions here (1200×900) are **identical** to the original `20260709_memory_vs_wm_v1` — the manipulation, in both this experiment and `20260714_memory_vs_wm_dense_reposition`, is doubling the *creature count* (5→10) while holding world size and food-object counts fixed, which halves food availability per creature relative to the original. An earlier report's wording ("2× the original world") described this loosely; this report uses "dense" to mean population density specifically, not world area.

Assumptions

- World layout: 1200×900, 10 creatures per trial, 500 RED_APPLE + 500 GREEN_APPLE + 500 GRAY_APPLE, 50 CACTUS, 100 ALOE — identical to `20260714_memory_vs_wm_dense_reposition` except `reposition = false`: eaten food does **not** respawn, so the world's food supply depletes over the run, same scarcity regime as `20260709_memory_vs_wm_v1`.
- `maxRuntimeMinutes = 60` for every condition.
- The `unified_critic` JEPA model represents the species prior for the WORLD_MODEL filter and the JEPA RPE baseline, same as in both prior experiments.

- All five conditions share the same world layout and creature count (n = 50 per condition: 5 trials × 10 creatures).

Conditions

Identical to `20260714_memory_vs_wm_dense_reposition`'s condition ladder — see that report for the full per-step rationale. Summary:

#	Key	Filters	Consolidation	Expectancy
1	`1_baseline`	TARGET_DISTANCE, AFFORDANCE, RANDOM	off	DISCRETE
2	`2_memory_only`	+ MEMORY	off	DISCRETE
3	`3_memory_consolidation`	+ MEMORY	**on**	DISCRETE
4	`4_jepa_rpe_only`	TARGET_DISTANCE, AFFORDANCE, WORLD_MODEL, RANDOM	off	**JEPA**
5	`5_jepa_rpe_consolidation`	TARGET_DISTANCE, AFFORDANCE, WORLD_MODEL, RANDOM	**on**	**JEPA**

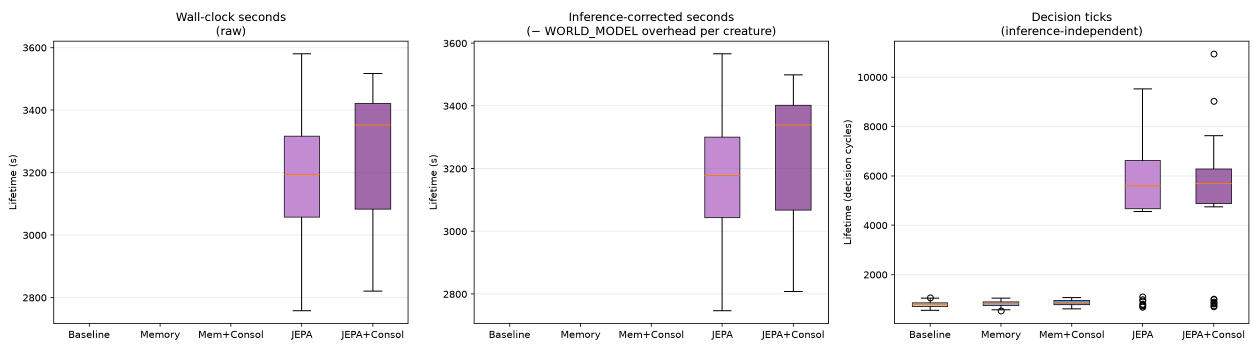
Hypothesis

#	Hypothesis
H1	Under scarcity (reposition=false), the JEPA survival advantage seen in the original scarce-world experiment returns at this larger world/population scale
H2	The episodic memory filter's survival penalty (seen in the original scarce-world experiment) also returns under scarcity
H3	Memory consolidation improves on memory-only performance under scarcity
H4	JEPA-adapter consolidation improves on JEPA-only performance under scarcity

#	Hypothesis
H5	JEPA’s Tedium suppression (present under scarcity originally, absent under abundance) returns

Results

1. Survival



Lifespan

Condition	Deaths	Mean lifetime (s)	± SD	Mean decision ticks	± SD
Baseline	**0/50**	— (no deaths)	—	787	109
Memory	**0/50**	— (no deaths)	—	826	113
Mem+Consol	**0/50**	— (no deaths)	—	864	105
JEPA	**24/50**	3212	213	5061	2313
JEPA+Consol	**17/50**	3241	231	5028	2322

The result is the **opposite** of H1: baseline, Memory, and Mem+Consol have **zero deaths** — every one of 150 creatures across those three conditions survived the full 60-minute cap. Both JEPA conditions show substantial death rates instead (48% and 34%), with deaths spread across roughly the second half of the run (mean lifetime ~3200s out of a 3600s cap, SD ~215-230s).

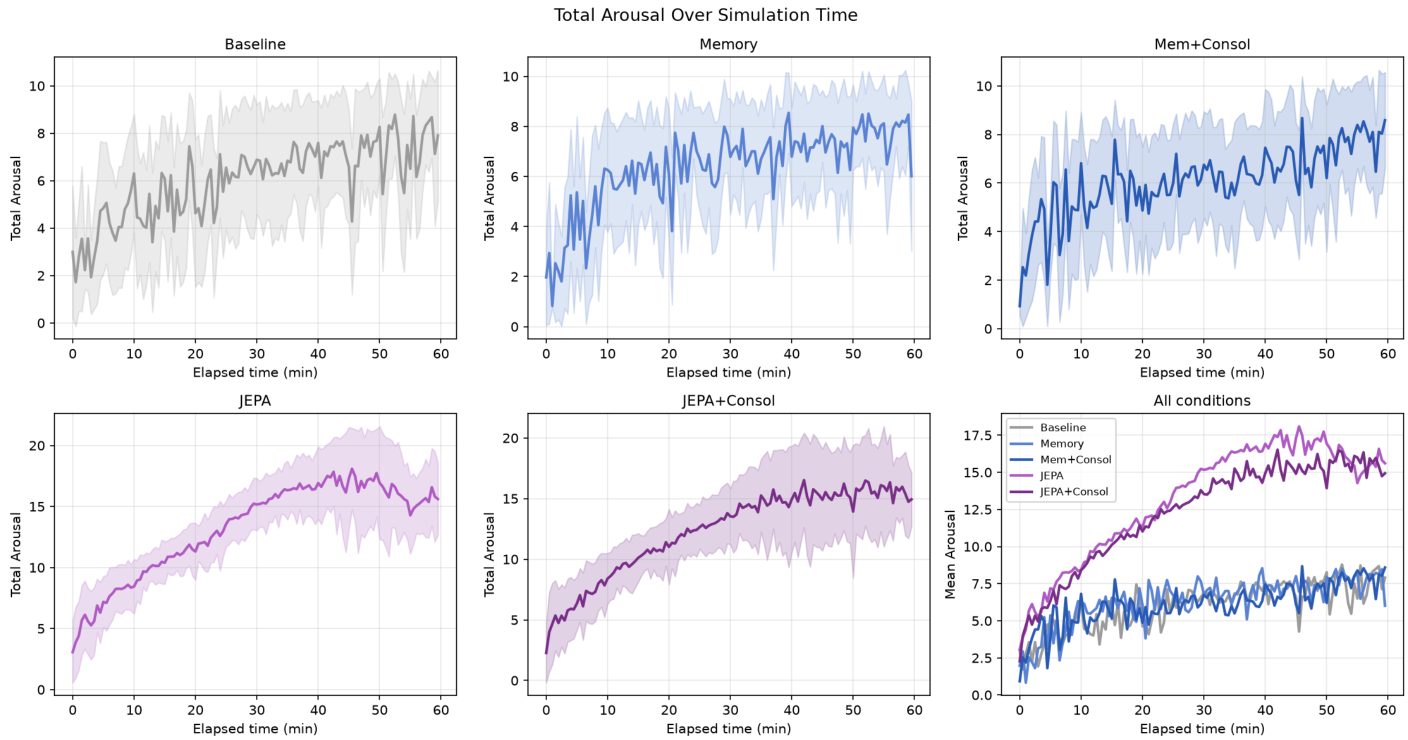
Decision-tick counts (Kruskal-Wallis H=123.8, p<0.0001) show baseline/Memory/Mem+Consol clustered tightly around 787-864 total ticks for the entire 60-minute life — JEPA and JEPA+Consol average **~6× that** (5028-5061 ticks, all pairwise comparisons vs. non-JEPA conditions p<0.0001). This tick-count gap is the first clue to what’s actually happening (see Analysis).

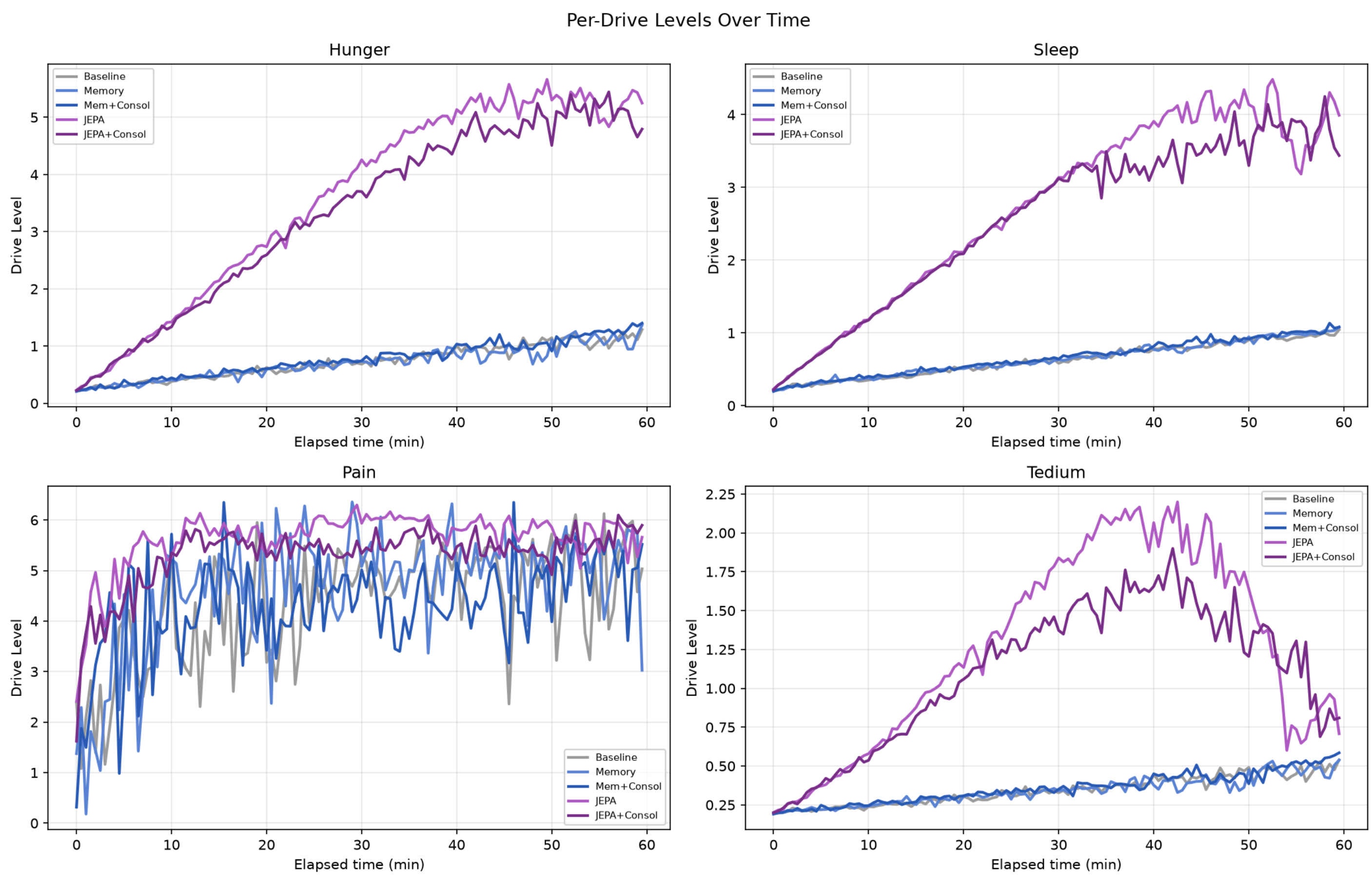
> **H1: Not confirmed — inverted.** JEPA does not gain a survival advantage under this scarcity configuration; it suffers a severe survival *penalty* relative to baseline, which is the exact

> opposite of the original `20260709_memory_vs_wm_v1` result (JEPA 720s vs. baseline 290s there).

> **H2: Not confirmed.** Memory shows no survival penalty here (tied with baseline at 0/50 deaths), unlike the original experiment where Memory conditions had shorter raw lifetimes than baseline (237-261s vs. 290s, both significant).

2. Drive Regulation





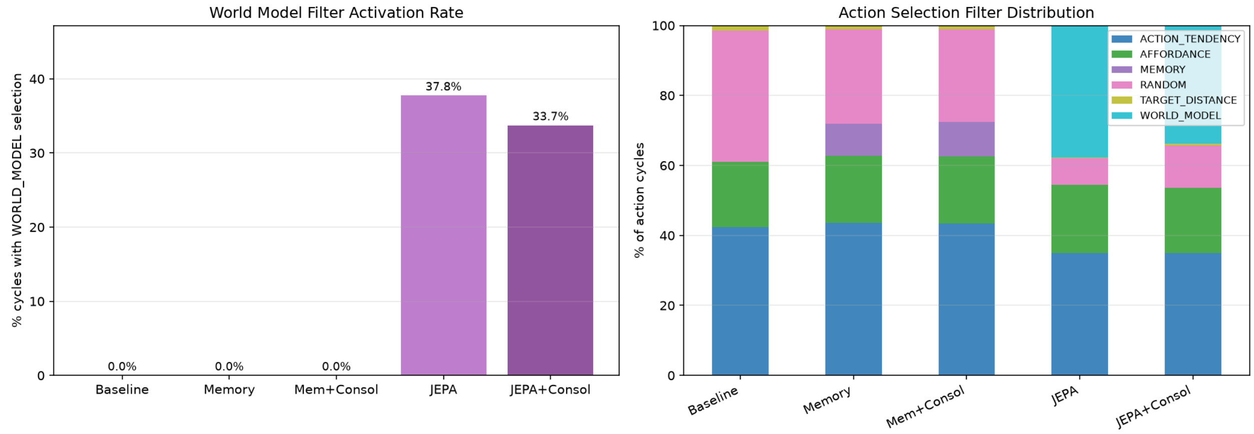
Metric	Baseline	Memory	Mem+Consol	JEPA	JEPA+Consol
Mean arousal	6.15	6.68	6.33	**13.63**	12.98

Metric	Baseline	Memory	Mem+Consol	JEPA	JEPA+Consol
Mean Hunger	0.75	0.76	0.81	**3.79**	3.63
Mean Sleep	0.65	0.69	0.69	**2.88**	2.78
Mean Pain	4.41	4.88	4.46	**5.69**	5.41
Mean Tedium	0.35	0.36	0.37	**1.27**	1.16
Mean max hunger (per creature)	1.11	1.19	1.26	**5.56**	5.34
% creatures hitting hunger ≥6.9 (of 7 max)	0%	0%	0%	**52%**	36%

The per-drive trajectory plot is unambiguous: Hunger and Sleep both climb steadily and near-linearly for both JEPA conditions across the full 60 minutes (reaching ~5 and ~4 respectively), while staying nearly flat (under 1.5) for the three non-JEPA conditions. Pain is also consistently higher and Tedium peaks over 2× higher for JEPA. Every homeostatic drive that matters for survival is elevated simultaneously in JEPA creatures — not just hunger.

> **H5: Not confirmed — inverted.** Tedium is *higher* under JEPA here (1.16-1.27 vs. 0.35-0.37 for non-JEPA, a >3× gap), not suppressed as it was in the original scarce-world experiment (JEPA Tedium 0.74-0.82 vs. baseline 2.33-2.43 there — the JEPA advantage flips direction between the two experiments, consistent with everything else in this report).

3. Action Selection

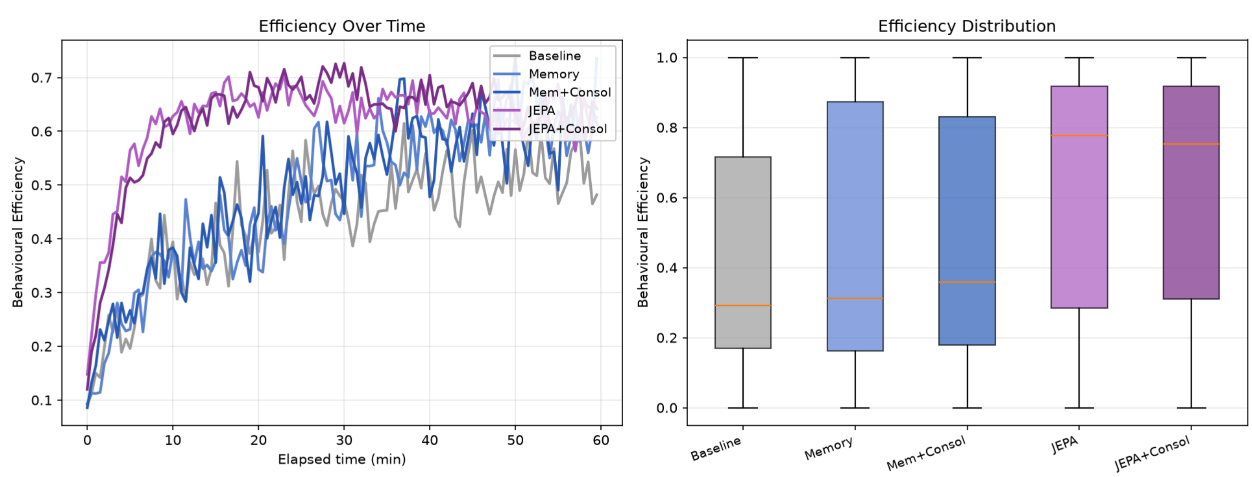


Filter distribution

Condition	ACTION_TENDENCY	AFFORDANCE	MEMORY	WORLD_MODEL	RANDOM
Baseline	42.3%	18.7%	—	—	37.5%
Memory	43.5%	19.2%	9.2%	—	27.1%
Mem+Consol	43.3%	19.3%	9.8%	—	26.5%
JEPA	34.9%	19.5%	—	**37.8%**	7.6%
JEPA+Consol	34.9%	18.7%	—	**33.7%**	12.1%

WORLD_MODEL fires on roughly a third of JEPA-condition decision cycles — high, consistent with the dense-reposition experiment’s finding that a denser world gives the filter more nearby candidates to evaluate per cycle.

4. Behavioural Efficiency



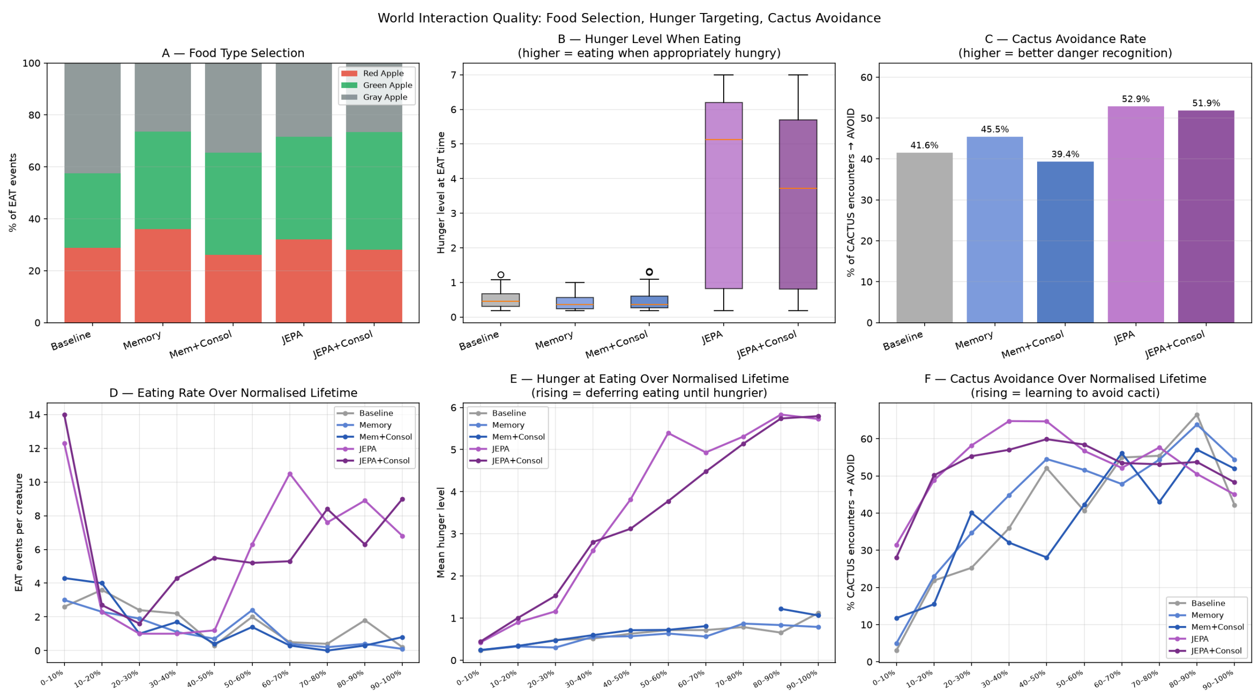
Efficiency

Condition	Mean efficiency
Baseline	0.42

Condition	Mean efficiency
Memory	0.46
Mem+Consol	0.47
JEPA	**0.61**
JEPA+Consol	**0.62**

JEPA conditions show notably *higher* per-action efficiency than non-JEPA — the opposite of what the survival numbers alone would suggest, and a piece of evidence that JEPA creatures are not executing actions poorly; each individual action is more purposeful. The problem, per Section 5 below, is what they choose to spend that purposeful action budget doing.

5. Eating Behaviour & Cactus Avoidance



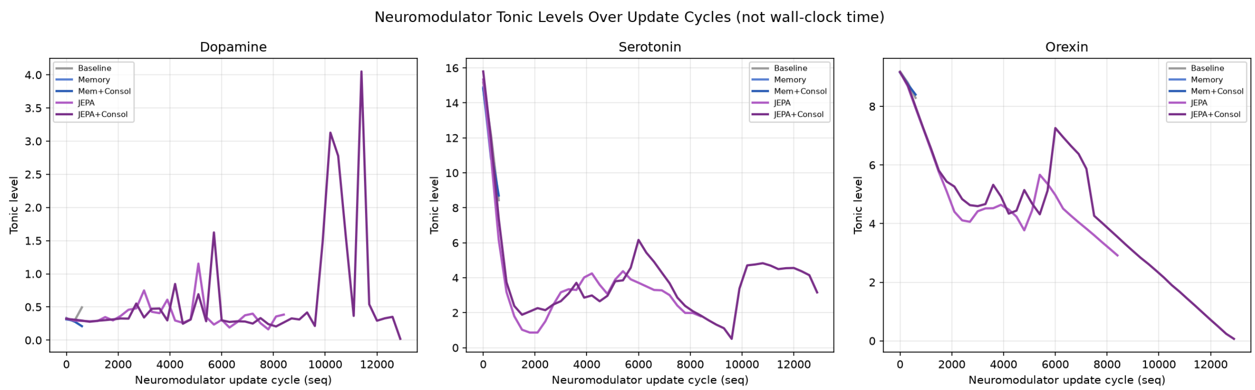
Eating behaviour and cactus avoidance

Condition	Cactus avoidance	Hunger at EAT (mean ± SD)	EAT events (5 trials total)
Baseline	41.6%	0.487 ± 0.238	160
Memory	45.5%	0.432 ± 0.227	125
Mem+Consol	39.4%	0.468 ± 0.263	142
JEPA	**52.9%**	**4.016 ± 2.507**	579
JEPA+Consol	**51.9%**	**3.457 ± 2.281**	623

JEPA conditions show *better* cactus avoidance (52.9%/51.9% vs. 39.4-45.5% for non-JEPA) — they are not simply worse at hazard recognition. But hunger-at-eating is dramatically higher for JEPA (~3.5-4.0 vs. ~0.43-0.49): JEPA creatures wait far longer, and get far hungrier, before eating.

Panel E (hunger-at-eating over normalised lifetime) makes the mechanism visible directly: JEPA's hunger-at-eating rises steadily from ~0.4 in the first life-decile to ~5.8 by the last — creatures progressively defer eating as their life goes on. Baseline's line stays flat at 0.3-0.8 across the *entire* lifetime, eating proactively and consistently regardless of how long they've survived. Panel D shows the same story from the supply side: JEPA/JEPA+Consol eat heavily in the first 10% of life (12-14 events/creature) then drop off sharply, while baseline maintains a low, steady eating cadence throughout.

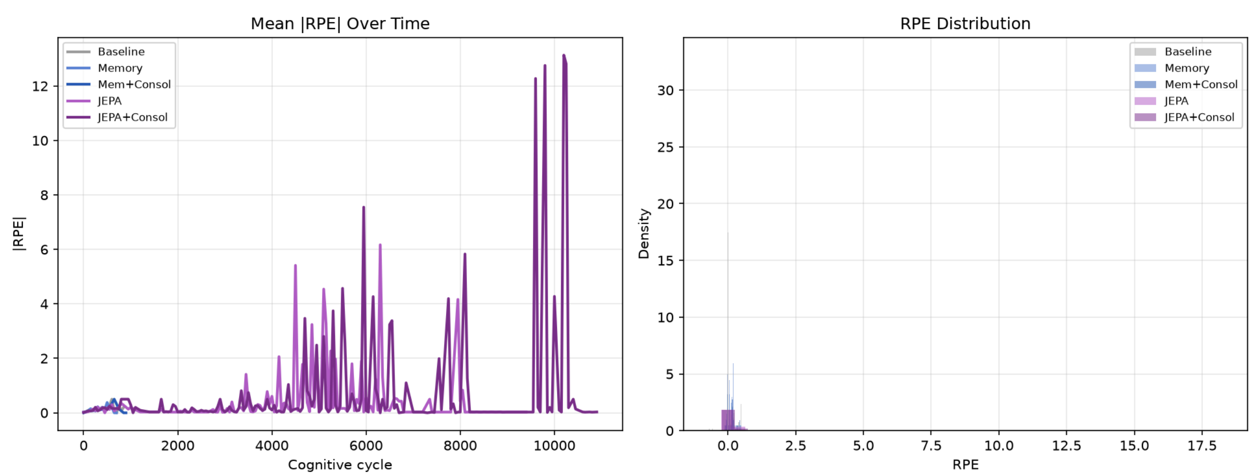
6. Neuromodulators



Neuromodulators

Not directly comparable across conditions on a shared x-axis — see the dense-reposition report's Section 8 for why (`seq` is a per-creature cycle counter, not wall-clock time, and JEPA's cycle rate differs from non-JEPA's).

7. Expectancy / RPE

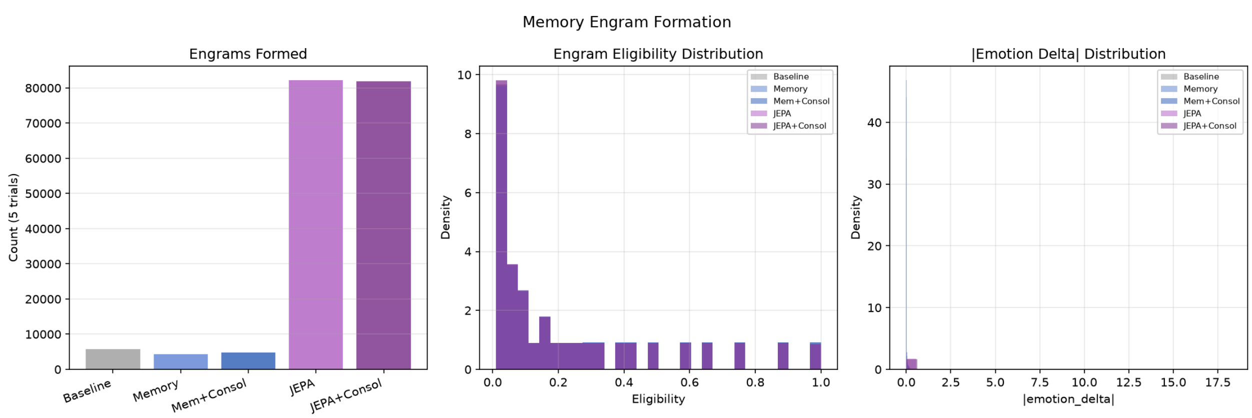


RPE

Condition	$\lvert RPE \rvert$ mean	SD
Baseline	0.1035	0.172
Memory	0.0975	0.137
Mem+Consol	0.0846	0.137
JEPA	**0.3597**	1.895
JEPA+Consol	**0.3260**	1.726

JEPA’s RPE remains clearly elevated (~3.5-4.3× baseline) — directionally consistent with both prior experiments, and further evidence the JEPA mechanism itself is functioning correctly; the survival cost comes from what the resulting novelty-seeking behavior does under real scarcity, not from a broken signal.

8. Memory Engrams

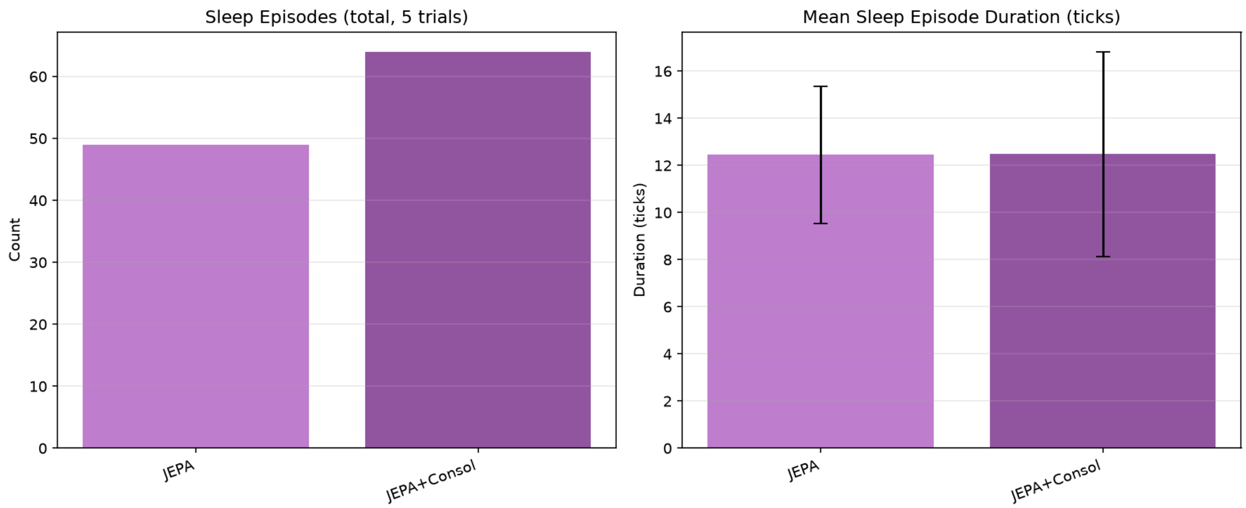


Engrams

Condition	Engrams (5 trials)	Mean $\lvert$delta$\rvert$
Baseline	5,719	0.0238
Memory	4,302	0.0224
Mem+Consol	4,773	0.0195
JEPA	**82,183**	**0.0806**
JEPA+Consol	**81,971**	0.0726

JEPA conditions form ~17× more engrams than non-JEPA — directly reflecting their ~6× higher tick count (more evaluation cycles → more engram-writing opportunities, same mechanism documented in the dense-reposition report’s Section 10) compounded with elevated engram salience from the larger RPE signal.

9. Sleep Episodes



Sleep

Condition	Episodes (5 trials)	Mean duration (ticks)
Baseline, Memory, Mem+Consol	0	—
JEPA	49	12.4
JEPA+Consol	64	12.5

Non-JEPA conditions show **zero sleep episodes across all 15 trials** — directly consistent with their flat, near-zero Sleep drive (Section 2): creatures that never accumulate meaningful sleep pressure never trigger sleep at all. JEPA+Consol shows more episodes than JEPA-only (64 vs. 49), mirroring the same pattern seen in the dense-reposition experiment (consolidation increases sleep engagement, independent of survival outcome).

10. JEPA Inference Latency

Condition	Count	Mean (ms)	Median (ms)
JEPA	95,557	6.51	5
JEPA+Consol	84,824	6.66	5

Confirms negligible inference overhead on CCAD, as expected — this experiment's entire point was to remove local-hardware inference latency as a confound, and it did.

Analysis

The central finding: under this population/food ratio, novelty-seeking is maladaptive

Every metric tells the same, internally consistent story:

1. **Baseline creatures barely act.** ~787-864 total decision ticks across a full 60-minute life — roughly one decision every 4-5 seconds. Hunger, sleep, pain, and tedium all stay low and flat the entire time. Every one of 150 creatures survives to the time cap.
2. **JEPA creatures act constantly.** ~6× the decision ticks, driven by a large, sustained RPE signal (Section 7) that keeps novelty-seeking behavior firing continuously. This is the JEPA mechanism working exactly as designed — the same elevated RPE/engram-salience signature documented in both prior experiments.
3. **That constant activity is not poorly executed** (Section 4: JEPA's per-action efficiency is *higher* than baseline's, not lower) **and is not reckless around hazards** (Section 5: JEPA's cactus avoidance is *better* than baseline's, not worse).
4. **But it systematically deprioritizes eating.** JEPA creatures defer eating until they are already critically hungry (hunger-at-eating climbing to ~5.8/7 by end of life, vs. baseline's flat ~0.3-0.8), and every homeostatic drive — not just hunger — climbs simultaneously and stays elevated. This is not an eating-avoidance bug specifically; it looks like a creature whose action selection is dominated by "what's interesting" (novelty/prediction-error) rather than "what do I currently need," across the board.
5. **The two failure modes compound under real scarcity.** In `20260709_memory_vs_wm_v1` (5 creatures, same food supply, same world), this same novelty-driven behavior pattern was a winning strategy — better foraging quality outweighed the activity cost, because food was plentiful enough per-capita that constant exploration still reliably found meals. Here, with the population doubled against an unchanged food supply, the food-per-creature ratio is halved, and the exploration cost (time and energy spent chasing novelty instead of eating) apparently exceeds what the environment can support — the same behavioral profile that won under lighter scarcity now starves creatures under heavier scarcity.

This is a genuinely different, and arguably more informative, result than a simple "H1 confirmed" would have been: it demonstrates that JEPA's survival advantage under scarcity is not a uniform property of the mechanism, but is contingent on the specific scarcity level — reversing sign somewhere between this experiment's population density and the original's.

Why baseline is so inactive

Baseline's very low tick count (787, vs. JEPA's 5061) is itself worth flagging as a secondary observation, independent of the scarcity story: `PartialAppraisal`'s cognitive cycle rate is event-driven, fired by perceptual events forwarded from `SensoryCortex` (see the dense-reposition report's neuromodulator-axis discussion for the mechanism). A creature that finds food nearby quickly and doesn't need to explore generates far fewer novel perceptual events than one constantly moving through the world — baseline's TARGET_DISTANCE/AFFORDANCE-driven "go to nearest known-good thing" strategy is, by construction, low-activity when the environment cooperates. Under this experiment's per-capita food density, it evidently cooperates enough that baseline creatures rarely need to move far or often.

What this says about `consolidationEnabled`

Both consolidated conditions (Mem+Consol, JEPA+Consol) track their non-consolidated counterparts closely on every survival/behavioral metric — Mem+Consol shows 0/50 deaths exactly like Memory, and JEPA+Consol's death rate (17/50) is numerically *lower* than JEPA-only's (24/50), though the survival-time distributions are not significantly different (p=0.72). Consolidation's one clear, consistent effect across both this and the dense-reposition experiment is increased sleep engagement (64 vs. 49 episodes here), not a change in survival outcome.

> **H3: Not confirmed** (both at 0/50 deaths, no meaningful difference to measure). **H4: Not confirmed** on lifetime (p=0.72 ns), though JEPA+Consol's numerically lower death count (17/50 vs. 24/50) is a small, non-significant signal in the *expected* direction — worth a larger follow-up sample if this specific question (does consolidation help JEPA cope with scarcity) is of interest.

Methodology Note: infrastructure issues encountered (non-scientific)

This experiment's data collection surfaced three real, now-fixed infrastructure bugs, none of which affected the validity of the 25 trials' final data (each is either fully independent of the scientific content or was caught and corrected before any trial's real data was lost):

1. **Empty-world image bug.** The first submission attempt ran against a stale CCAD image built before this experiment's simulation configs existed in the merged codebase — `docker/Dockerfile` bakes `simulations/*.conf` into the image at build time, and CCAD only pulls prebuilt images from GHCR. Caught within minutes via a direct log check (`worldObjectSettings.size=0`); resolved by merging the pending PR and switching the CCAD image pin back to CI-tracked `:latest`.
2. **`manifest.json` concurrency race.** Up to 25 concurrent `extract.py` processes (one per SLURM array task) read-modify-wrote a single shared `manifest.json` file with no locking, corrupting it under concurrent writes and crashing extraction for 14/25 trials — but only *after* each trial's real data (parquet + postgres backup) had already been written successfully. Fixed with a proper lock + atomic write (`scripts/dl2l_data/manifest.py`, commit `cd95c98`), verified with a 25-process concurrency stress test. One trial's ad-hoc single-array resubmission (to

backfill a separately-lost trial, see below) bypassed the normal sync step and hit the same bug against an already-corrupted remote file before the fix and a fresh manifest were in place.

3. **NFS quota exhaustion.** The preserved-overlay safety net (kept for recovering genuinely failed trials) accumulated ~19GB of leftover `pg_overlay.img` files from the 14 manifest-race failures, pushing the CCAD account against its NFS storage quota and causing one trial (`5_jepa_rpe_consolidation/trial_5`) to lose its data entirely (crashed on the very first parquet write, before any output existed). Resolved by cleaning up the overlay files and re-running that single trial in isolation once quota headroom was restored.

All 25 trials' final data was directly verified (file presence, creature counts, and — for this report specifically — real death-time and drive-trajectory variance) before analysis, independent of any `DONE` sentinel value.

Summary Table

Metric	Baseline	Memory	Mem+Consol	JEPA	JEPA+Consol
Deaths	0/50	0/50	0/50	**24/50**	17/50
Mean lifetime (s)	—	—	—	3212	3241
Mean decision ticks	787	826	864	**5061**	5028
Mean max hunger	1.11	1.19	1.26	**5.56**	5.34
Mean arousal	6.15	6.68	6.33	**13.63**	12.98
Mean Tedium	0.35	0.36	0.37	**1.27**	1.16
Mean efficiency	0.42	0.46	0.47	**0.61**	0.62
Cactus avoidance	41.6%	45.5%	39.4%	**52.9%**	51.9%
Hunger at EAT	0.49	0.43	0.47	**4.02**	3.46
Sleep episodes	0	0	0	49	**64**
IRPEI mean	0.104	0.098	0.085	**0.360**	0.326
Engrams (5 trials)	5,719	4,302	4,773	**82,183**	81,971
WORLD_MODEL %	0.0%	0.0%	0.0%	37.8%	**33.7%**

Conclusions

H1: Not confirmed — the JEPA survival advantage from the original scarce-world experiment does not just fail to return, it inverts. Baseline survives perfectly (0/50 deaths); JEPA suffers substantial mortality (24/50). This is the headline, unexpected result of this experiment.

H2: Not confirmed. Memory shows no survival disadvantage under this scarcity configuration (tied with baseline at 0/50 deaths), unlike the original experiment.

H3: Not confirmed. Memory and Mem+Consol are statistically indistinguishable (both 0/50 deaths).

H4: Not confirmed on survival-time distribution ($p=0.72$ ns), though JEPA+Consol's lower raw death count (17/50 vs. 24/50) is a small, non-significant trend worth a larger follow-up sample.

H5: Not confirmed — inverted. JEPA's Tedium is *higher* than baseline's here (1.16-1.27 vs. 0.35-0.37), the opposite of the strong suppression effect (0.74-0.82 vs. 2.33-2.43) seen in the original scarce-world experiment.

The overarching finding: **JEPA's novelty-driven action selection is not uniformly adaptive under scarcity — its value is itself contingent on exactly how scarce the environment is.** At the original experiment's population density (5 creatures, this same food supply), constant exploration paid for itself in better foraging outcomes. At this experiment's density (10 creatures, same food supply — half the food per creature), the same behavioral profile systematically deprioritizes eating relative to novelty-seeking, and creatures pay for it with their lives. Baseline's simple, reactive, low-activity strategy is comparatively far more robust to this specific stressor, even though (per the dense-reposition experiment) it shows no particular advantage — or disadvantage — once scarcity is removed entirely.

Next Steps

1. **Locate the reversal point.** Run this same condition ladder at one or two intermediate population densities (e.g., 6-8 creatures against the same fixed food supply) to find where JEPA's survival advantage flips to a penalty — this would directly test the "exploration cost exceeds what the environment can support" explanation above.

2. **Test a hunger-weighted action-selection modification for JEPA.** If JEPA's core issue is novelty dominating homeostatic need in action selection, a version of the WORLD_MODEL filter that down-weights RPE-driven candidates when hunger is already elevated would be a direct, falsifiable test of that mechanism.

3. **Larger sample for the consolidation question.** JEPA+Consol's lower death count than JEPA-only (17/50 vs. 24/50) didn't reach significance at $n=50$ per condition; a larger run focused specifically on this comparison could resolve whether consolidation provides a real, if modest, survival benefit under scarcity.

Data Availability

```
ml/data_20260717_memory_vs_wm_dense_scarce/   — conditions 1–5 (5 trials × 10 creatures each)
```

Pending upload to `felipedreis/dl2l-experiments` under prefix `20260717_memory_vs_wm_dense_scarce/` — blocked on HuggingFace re-authentication (the cached token is invalid; `hf auth login --force` needed before upload can proceed).